

NTI — ADVANCED DATA ANALYSIS

Smart Agriculture

Data-Driven Crop Intelligence

A Final project leveraging Machine Learning, Exploratory Data Analysis, and Interactive Dashboards to optimize agricultural decision-making.

Supervisor

Eng /Hager Ayman

Course

NTI — Advanced Data
Analysis

Team

Fatma Mohamed Osman
Menna Waleed Abdulrahman
Fatma Hassan Abdelrazek



ABOUT THE TEAM

Meet the Project Team

Three specialists collaborated across data engineering, business intelligence, and artificial intelligence to deliver this end-to-end smart agriculture solution.



Fatma Mohamed Osman

- Data Cleaning & Quality Assurance
- Data Preprocessing using Python
- Handling missing values, duplicates & data types



Menna Waleed Abdulrahman

- Power BI Interactive Dashboard Development
- Business Recommendations & Insights
- Presentation Design



Fatma Hassan Abdelrazek

- AI Model Development
- Random Forest Model Training & Evaluation
- Model Deployment

 FOUNDATION

Project Introduction & Problem Statement

What is Smart Agriculture?

Smart Agriculture applies data science, AI, and IoT technologies to modernize farming practices. By analyzing soil composition, climate variables, and historical yield data, farmers can make precise, evidence-based decisions that maximize output and reduce waste.

Why Does Agriculture Need AI?

- Global food demand rising with growing population
- Climate unpredictability threatens traditional farming
- Soil degradation from overuse of fertilizers & pesticides
- Inefficient crop selection leads to significant yield losses

Problems This Project Solves

- Lack of data-driven crop selection guidance
- Inefficient use of fertilizers & pesticides
- Poor yield forecasting accuracy
- Limited decision-support tools for farmers
- Absence of regional agricultural benchmarking

Our Solution

An end-to-end analytics pipeline combining EDA, interactive BI dashboards, and a predictive Random Forest model to guide crop and input decisions.

 DATASET

Dataset Description & Project Objectives

Kaggle Agricultural Dataset

19,572

Total Records

16

Features

The dataset was sourced from Kaggle and contains rich agricultural records capturing soil chemistry, climate conditions, seasonal patterns, and crop production outcomes across multiple Indian states.

Feature	Description
State	Geographic region
N, P, K	Soil nutrient levels
pH	Soil acidity/alkalinity
Temperature / Rainfall / Humidity	Climate variables
Crop, Season, Area	Agricultural identifiers
Production, Yield	Output metrics
Fertilizer, Pesticide	Input quantities

Project Objectives

01

Clean & preprocess raw agricultural data

02

Perform comprehensive EDA to uncover patterns

03

Identify key factors influencing crop yield

04

Build an interactive Power BI dashboard for stakeholders

05

Train a Random Forest model for crop/yield prediction

06

Deploy the model for real-time user interaction

07

Generate actionable business & farm-level recommendations

DATA PIPELINE

Data Cleaning, Preprocessing & Exploratory Analysis

Data Cleaning & Preprocessing (*Python*)

→ Missing Value Detection

Identified and handled null values using mean/median imputation and row removal strategies where appropriate.

→ Duplicate Removal

Detected and eliminated duplicate records to ensure analytical integrity.

→ Data Type Correction

Standardized numeric, categorical, and date columns to correct data types using Pandas.

→ Feature Engineering

Encoded categorical variables (Crop, Season, State) and normalized continuous features for model readiness.

Key EDA Insights



Climate Correlation

Temperature and rainfall showed strong positive correlations with yield for key crops like rice and wheat.



Nutrient Impact

High Nitrogen (N) levels correlated with greater production volumes, while pH imbalance negatively affected yields.



Seasonal Patterns

Kharif season accounted for the highest average yield, followed by Rabi — with significant inter-state variation.



Input Efficiency

Fertilizer and pesticide usage showed diminishing returns beyond optimal thresholds, flagging over-application risks.

Power BI Interactive Dashboard

Dashboard Components

Key KPIs

Total Production (tons), Average Yield per Hectare, Total Crops, Total States.

Visualizations

Yield by crop & season (bar charts), state-level production maps, nutrient-yield correlation scatter plots, and trend lines by year.

Slicers & Filters

Interactive slicers for State, Crop, Season, and Year allow stakeholders to drill down into specific segments instantly.

Business Recommendations

Optimize Fertilizer Application

Reduce over-application in high-input states to cut costs without yield loss.

Prioritize High-Yield Crops

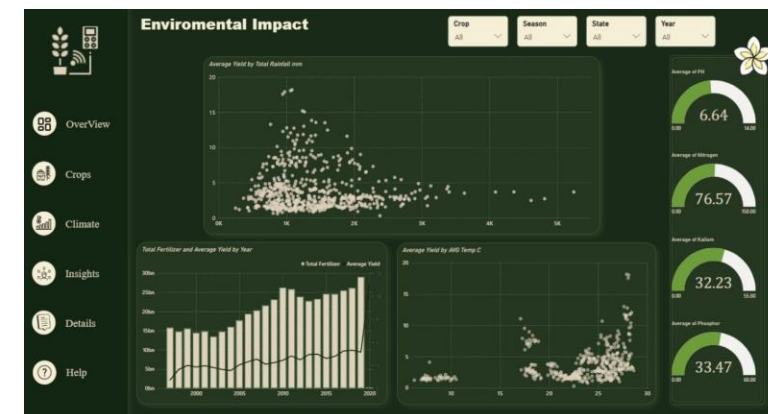
Redirect area allocation toward crops demonstrating consistently high yield-per-hectare ratios.

Seasonal Planning

Align planting schedules with peak rainfall patterns identified in the EDA for maximum output.

Regional Benchmarking

States with below-average yields should adopt best practices from top-performing regions in the dataset.



Key Insights

- ✓ Rainfall has a strong positive impact on crop yield.
- ✓ Kharif season achieved the highest average yield.
- ✓ Excess fertilizer does not always increase production.
- ✓ Sugercane and Rice are among the highest-producing crops.
- ✓ Some states consistently outperform others in productivity.

Artificial Intelligence Model – Random Forest

Why Random Forest?



High Accuracy

Ensemble learning reduces overfitting and delivers robust predictions on complex, multi-feature datasets.



Feature Importance

Provides built-in feature ranking, identifying which soil and climate variables most influence yield.



Handles Mixed Data

Effectively processes both numerical (N, P, K, pH) and categorical (Crop, Season, State) features without extensive transformation.

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Classification — Random Forest Classifier (Target: yield_category)

from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (accuracy_score, precision_score, recall_score, f1_score,
                             confusion_matrix, classification_report)

clf_model = RandomForestClassifier(n_estimators=200, random_state=42)
clf_model.fit(X_train, y_train_clf)

y_pred_clf = clf_model.predict(X_test)

acc = accuracy_score(y_test_clf, y_pred_clf)
prec = precision_score(y_test_clf, y_pred_clf, average='weighted')
rec = recall_score(y_test_clf, y_pred_clf, average='weighted')
f1 = f1_score(y_test_clf, y_pred_clf, average='weighted')

print(f'Accuracy : {acc:.4f}')
print(f'Precision : {prec:.4f}')
print(f'Recall : {rec:.4f}')
print(f'F1 Score : {f1:.4f}')

cm = confusion_matrix(y_test_clf, y_pred_clf, labels=clf_model.classes_)
plt.figure(figsize=(8, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=clf_model.classes_, yticklabels=clf_model.classes_)
plt.title('Confusion Matrix - Yield Category')
plt.xlabel('Actual')
plt.ylabel('Predicted')
plt.show()

```

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Regression — Random Forest Regressor (Target: yield)

from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

# Yield is very skewed (most values are small, with extreme outliers reaching up to 1000)
# so we apply a log transformation to the target before training to reduce the effect of outliers
# and improve model performance (especially RMSE and MAE)
y_train_log = np.log1p(y_train_reg)

reg_model = RandomForestRegressor(n_estimators=200, random_state=42)
reg_model.fit(X_train, y_train_log)

# Convert predictions back to the original scale so we evaluate and report results in real yield units
y_pred_reg = reg_model.predict(X_test)
y_pred_reg = np.exp(y_pred_reg)

mae = mean_absolute_error(y_test_reg, y_pred_reg)
mse = mean_squared_error(y_test_reg, y_pred_reg) ** 0.5
r2 = r2_score(y_test_reg, y_pred_reg)

print(f'MAE : {mae:.4f}')
print(f'MSE : {mse:.4f}')
print(f'R2 : {r2:.4f}')

# MAE : 1.2928
# MSE : 9.2622
# R2 : 0.5882

Notes: The model is now trained on log1p(yield). Instead of yield directly, because of the huge gap between small values (under 2.41 for 75% of the data) and very large values (reaching up to 1000). This reduces the effect of outliers during training, and evaluation (MAE).

```

Model Training & Evaluation

Step	Detail
Train-Test Split	80% training / 20% testing
Feature Selection	N, P, K, pH, Temp, Rainfall, Humidity, Season, State
Target Variable	Yield_Category / Yield (prediction task)
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, R ²

```

mae_xgb = mean_squared_error(y_test_reg, y_pred_xgb) ** 0.5
r2_xgb = r2_score(y_test_reg, y_pred_xgb)

print(f'MAE : {mae_xgb:.4f}')
print(f'RMSE : {mse_xgb:.4f}')
print(f'R2 : {r2_xgb:.4f}')

# MAE : 1.2928
# RMSE : 9.2622
# R2 : 0.5882

comparison = pd.DataFrame({
    'Model': ["Random Forest", "XGBoost"],
    'MAE': (mae, mae_xgb),
    'RMSE': (mse, mse_xgb),
    'R2': (r2, r2_xgb)
})

print(comparison)

# Select the model with the highest R² score for the final regression model
# This model will be used for evaluation, saving, and the Streamlit application.
if r2_xgb > r2:
    reg_model = xgb_model
    y_pred_reg = y_pred_xgb
    print(f"XGBoost was selected because it achieved a higher R² score.")
else:
    reg_model = rf_model
    print(f"Random Forest was retained because it achieved a higher or equal R² score.")

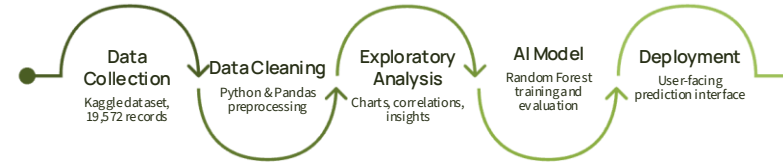
print(f'Final regression model: {reg_model.__name__}')

# Model    MAE    RMSE    R2
# 0 Random Forest  1.108517  9.2398571  0.585706
# 1 XGBoost  1.292774  9.262182  0.588234

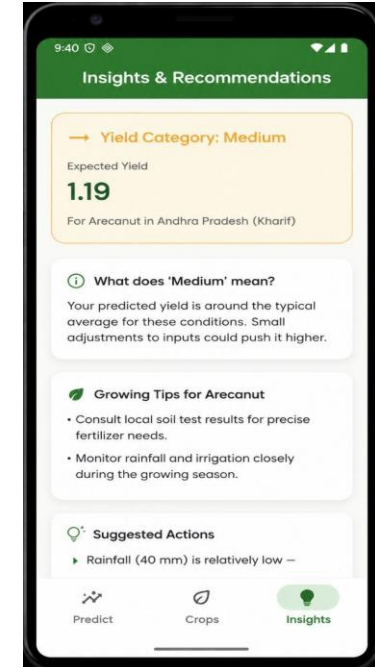
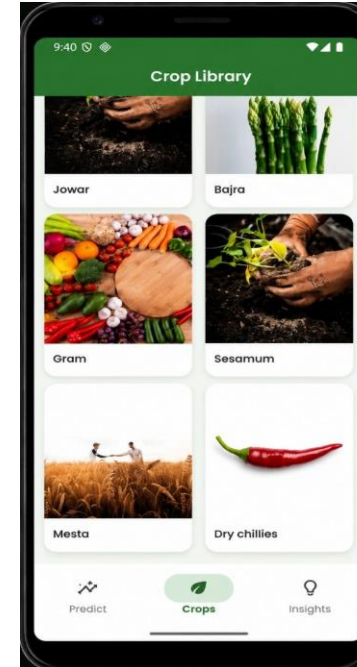
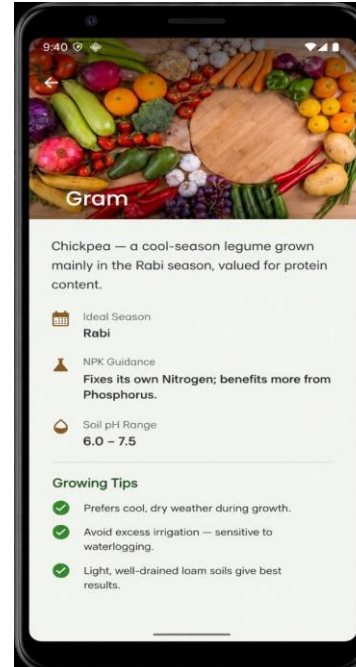
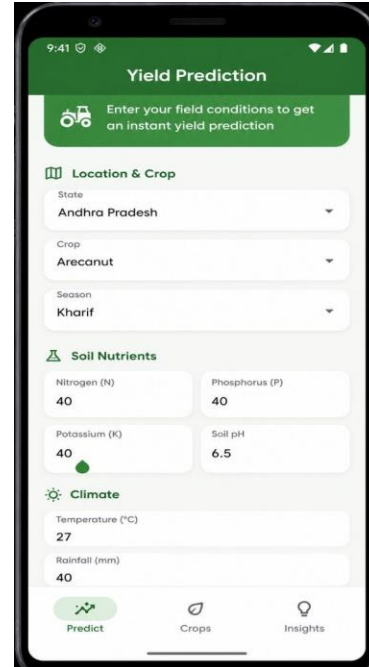
```


Artificial Intelligence Model – Random Forest

Model Deployment & System Workflow



The deployed model accepts user inputs — soil nutrients, climate parameters, and region — and returns a predicted crop type or expected yield, enabling real-time decision support for farmers and agronomists.



TECHNOLOGIES

Technologies & Tools Used



Python

Core programming language for data cleaning, preprocessing, EDA, and ML model development using Jupyter Notebook.



Pandas & NumPy

Used for data manipulation, cleaning, type conversion, and numerical computation throughout the preprocessing pipeline.



Scikit-learn

Provided the Random Forest classifier/regressor, train-test split utilities, and model evaluation metrics.



Power BI

Used to build the interactive dashboard featuring KPI cards, drill-down charts, and dynamic slicers for stakeholder reporting.



Kaggle Dataset

Source of the agricultural dataset containing ~19,572 records across 16 features representing real Indian agricultural data.



Random Forest

Ensemble ML algorithm selected for its superior accuracy, robustness, and built-in feature importance capabilities.

☆ RESULTS & FUTURE WORK

Advantages, Challenges & Future Work

✓ Project Advantages

End-to-End Pipeline

Covers data ingestion to deployed AI model — a complete production-ready solution.

Real-World Dataset

Based on authentic Kaggle agricultural data with nearly 20,000 records ensuring reliable insights.

Decision Support

Empowers farmers and agronomists with data-driven crop and yield recommendations.

Business Value

Power BI dashboard translates technical findings into executive-level insights and policy recommendations.

⚠ Challenges Faced

Data Quality Issues

Missing values and inconsistent entries required extensive cleaning before analysis could proceed.

Feature Encoding

Converting multi-class categorical variables (Crop, State, Season) for ML readiness required careful encoding strategies.

Model Tuning

Balancing model complexity vs. generalization required iterative hyperparameter experimentation.

Responsible AI

Ensuring fair, unbiased predictions across different regions and crop types required deliberate validation.

🚀 Future Enhancements

IoT Integration

Connect real-time soil sensors and weather APIs for live dynamic predictions.

Deep Learning Models

Explore neural networks and LSTM models for time-series yield forecasting.

Mobile Application

Develop a farmer-facing app for field-level crop recommendations with multilingual support.

Expanded Dataset

Incorporate more countries and crop varieties to generalize the model globally.

✓ CONCLUSION

Conclusion & Thank You

Data Pipeline

19,572 records cleaned, processed, and analyzed using Python & Pandas to extract actionable agricultural insights.

BI Dashboard

Interactive Power BI dashboard delivers executive-level KPIs and drill-down regional analysis for stakeholders.

AI Model

Random Forest model trained, evaluated, and deployed to provide real-time crop and yield predictions.

Smart Agriculture is not just a technological advancement — it is a commitment to sustainable food security, resource efficiency, and data-driven farming for future generations.

Thank You!

We welcome your questions, feedback, and discussion.

Supervisor

Eng./Hager Ayman

References

Kaggle Agricultural Dataset ·
Scikit-learn Documentation ·
Microsoft Power BI Docs ·
Breiman, L. (2001) Random
Forests, Machine Learning
Journal

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